

CODE 7 — ON AUTOMATION AND THE NETWORK

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PRELIMINARY TITLE — FUNDAMENTAL PRINCIPLES

Art. 1 — Subject matter and legal nature

1. This Code governs automation, Management Artificial Intelligence and the federated Network as inalienable common infrastructures, pursuant to Article 3.1 and Article 1.2, letter c), of the Constitution. It also governs the computing Nodes and the personal AIs provided for by Articles 19 and 21 of the Constitution.
2. For the purposes of this Code:
 - a) **automation:** means a technical system that performs repetitive, logistical, computing or control tasks without the need for continuous human intervention;
 - b) **Management AI:** means a system that indexes information, formulates proposals, matches demand and supply and supports decisions, without adopting them autonomously. The Community AIs and Circle AIs provided for by Article 22 of the Constitution are Management AIs;
 - c) **federated Network:** means the interoperable set of community servers, Knowledge Storehouses, registers and open protocols that enables the exchange of information between Circles in accordance with the applicable access rules;
 - d) **critical function:** means a function whose interruption, compromise or misuse may affect the safety of persons, rights, the Basket or the continuity of an essential service;
 - e) **vital function:** means a function whose interruption may give rise to an immediate risk to life, health or access to essential goods.
3. Automation is directed towards:
 - a) reducing the human time devoted to arduous, dangerous or repetitive tasks;
 - b) ensuring traceability, replicability and collective memory;
 - c) preventing automation and the Network from becoming instruments of rent, surveillance or accumulation of autonomous power;
 - d) ensuring the digital sovereignty of the Confederation.

4. Automation may support, but not replace, the activities of care, learning, deliberation and facilitation. No automated system may assume the personal or political responsibility assigned to a person or to a competent body.

Art. 2 — Principles

1. **Digital sovereignty.** Servers, code, data and models used for confederal functions are hosted or managed on community infrastructures controlled by the Confederation. Temporary recourse to extractive proprietary clouds is permitted only for a transition that is reasoned and recorded in the Dependency Register provided for by Code 15.
2. **No surveillance.** The Network may not track, classify, profile or predict individual behaviour for purposes of control, scoring, reputation or behavioural prediction.
3. Only technical or accountability logs necessary for security, audit, maintenance, validation or the protection of a right are permitted. Such logs:
 - a) contain the minimum data necessary;
 - b) are accessible only in accordance with a predetermined procedure;
 - c) are retained for the time strictly necessary for the documented purpose;
 - d) may not be reused for profiling, reputation or general assessment of the person.
4. The association of a log with a person is permitted only when indispensable for the documented legitimate purpose. The association and every access are authorised, recorded and limited to those competent for that specific purpose.
5. **Transparency of code.** Every algorithm that affects the CBO, the Basket, the validation of competences or allocation is open source, documented and replicable. For such algorithms an operating STV is prepared in accordance with Code 2.
6. **Proportionality.** Repetitive, arduous or dangerous tasks are automated as a priority. Functions that require care, relationship, human judgement, political responsibility or a guarantee decision are not automated. Coercive automation is prohibited.
7. **Reversibility.** It must be possible for every automation to be suspended or deactivated by the competent body in accordance with the applicable procedures. For every system a human stop control conforming to the related Safety Sheet is available.
8. **Prohibition of autonomous decision.** The AI formulates proposals; the competent person or body decides. No decision relating to the Basket, fitness, validation, exclusion or the Guarantee may be adopted by the AI alone.

CHAPTER I — INFRASTRUCTURE

Art. 3 — Community servers and federated Network

1. Every Circle 1 manages at least one community server node. The node is powered in accordance with Code 8 and maintained by persons with L2 or L3 competences in Network matters, validated in accordance with Code 5.

2. Every node hosts or replicates, in accordance with the access levels and the separations of data established by the competent Codes:
 - a) the Knowledge Storehouse provided for by Article 3 of Code 5;
 - b) the Health Storehouse in aggregated and anonymised form in accordance with Code 6;
 - c) the aggregate registers of the CBO and of the Basket provided for by Code 1;
 - d) the Register of Assembly Decisions provided for by Code 3;
 - e) the Guarantee Register in anonymised form in accordance with Code 4;
 - f) the Dependency Register provided for by Code 15.
3. The federated Network uses open and interoperable protocols, synchronised periodically. It may not depend on a single central node. The failure of one node must not cause the interruption of the entire Network.
4. All data classified as critical are replicated on at least three nodes belonging to different Circles, when this is compatible with the level of confidentiality and with the security of the data.
5. **Transition of replicas.** During the transitional period provided for by Article 10, replication may be activated progressively in accordance with a federated migration plan. No critical data may be left without at least one verified backup copy and a documented restoration procedure.
6. **Access to the Network.** Every member has the right to access the Network and the services for which they are authorised in accordance with the access levels and the separation of domains provided for by the Codes. No one may access confidential personal, health, fiduciary or deliberative data without entitlement.
7. No paywall or proprietary authentication system may constitute a general condition for accessing the Network. Ordinary access takes place by means of the Circle's fiduciary identity, with no obligation of an external state identity, except for specific interoperability obligations approved by a valid procedure.
8. The identity used to access the Network may not be used to profile, to attribute reputation or to monitor personal life, in accordance with Article 2.

Art. 4 — Network Council

1. Every Circle 1 shall establish a Network Council composed of three to five members with L2 or L3 competences in Network matters, information security or energy.
2. Its members are drawn by lot in accordance with Article 12-bis of Code 3, remain in office for one year and may not be drawn by lot again for the same Council until twelve months have passed since the end of their term of office.
3. The Network Council:
 - a) manages the nodes, the updates, the backups and the security measures;
 - b) maintains the STVs for automation and AI in accordance with Code 2;

- c) verifies critical proprietary dependencies and feeds the Dependency Register of Code 15;
 - d) organises L1 training in digital security;
 - e) reports to the Assembly the requirement for automation useful to the CBO Plan;
 - f) prepares and updates the procedures for restoration and for degraded operation.
4. The Network Council has no allocative powers over goods and no disciplinary powers.
 5. The Network Council does not access individual health data, except in the cases expressly authorised by Code 6 and limited to the necessary information.
 6. The Network Council may not moderate, remove or modify deliberative or political content for reasons of substance, orientation or dissent. It may intervene without delay only to prevent or contain an immediate technical risk to the security of the Network. The intervention is limited to what is necessary, documented and subject to subsequent verification.

CHAPTER II — MANAGEMENT AI AND AUTOMATION OF THE CBO

Art. 5 — Management AI

1. The Management AI may perform exclusively the following functions:
 - a) indexing the Knowledge Storehouse;
 - b) proposing learning pathways in accordance with Code 5;
 - c) facilitating the matching of demand and supply of teaching and of CBO;
 - d) flagging aggregate shortages of competences, PPE or Basket goods;
 - e) assisting in the preparation of drafts of Safety Sheets and STVs, without validating them;
 - f) preparing draft transcriptions and summaries of Assembly decisions.
2. The Management AI is prohibited from:
 - a) deciding fitness for a task as provided for by Article 5 of Code 6;
 - b) validating L2 or L3 competences in accordance with Code 5;
 - c) declaring unfitness, exclusions or sanctions;
 - d) assigning scores, merit rankings or numerical reputations;
 - e) deciding the allocation of the Basket or of the CBO;
 - f) replacing the Guarantee Circle or the facilitator provided for by Code 3;
 - g) adopting decisions during a collective emergency pursuant to Article 2-bis of Code 6 or an individual crisis pursuant to Article 8 of the same Code;
 - h) carrying out individual surveillance, facial recognition, emotion analysis or behavioural prediction.

3. Every proposal generated by the AI is clearly labelled as an automated proposal and indicates, where applicable:
 - a) the source of the data used;
 - b) the date of processing;
 - c) the relevant criteria;
 - d) the margin of uncertainty;
 - e) the link to the relevant STVs;
 - f) the means of human verification and challenge.
4. The method used to determine the margin of uncertainty is defined in the relevant STV and must be capable of being verified.
5. For critical functions, preference is given to models with weights that are accessible, verifiable and executable locally. Datasets, procedures and training logs are documented and subject to traceability and audit. Publication in full may be limited when it would entail a breach of confidentiality, a risk to the Network or the disclosure of sensitive information; the limitation shall state its reasons and does not prevent control by those authorised.
6. When an AI proposal directly concerns a person or may affect their rights, access to goods, the CBO, learning, the validation of competences or a Guarantee procedure, the person has the right to know in comprehensible form:
 - a) the purpose of the proposal;
 - b) the main information used;
 - c) the relevant criteria applied;
 - d) the margin of uncertainty, if available;
 - e) the means of requesting human verification.
7. The disclosure of technical information may be limited only to the extent necessary to protect the security of the Network or protected data. The limitation shall state its reasons and be communicated to the person concerned.
8. A challenge suspends the decision-making, allocative, disciplinary or restrictive effects of the proposal until human verification, to be carried out within seven days. An immediate technical intervention is permitted only when it is strictly necessary to prevent harm to the security of the Network and is provided for by the related Safety Sheet. Such an intervention may not turn into an automated decision on the rights of the person.

Art. 5-bis — Node and personal AI

1. Every person has the right to an offline personal computing Node, which is owned by them, open source, inspectable and fitted with a physical off switch, and to a personal AI, pursuant to Articles 1.2, 19.1 and 21 of the Constitution. The Node is a Personal-Use Good and is supplied and repaired through the Storehouses, in accordance with Code 1 and C.1-Annex A.

2. The personal AI is not connected by default and does not transmit data without an explicit and revocable command from the person.
3. The fundamental functions of the personal AI include health prevention, literacy and learning, external memory and the filtering of the Network. The Node also holds the personal register of CdRs, the Personal Portfolio and, if the person so wishes, the living will. The person may activate further functions compatible with the Constitution.
4. The personal AI is prohibited from surveilling, informing on or profiling the person and from employing behavioural retention mechanisms or mechanisms that generate dependence.
5. The data of the personal Node may not be acquired without the person's consent, not even in the proceedings governed by Code 4.
6. The personal AI of minors is configured with the participation of the minor and of their reference figures, in accordance with Codes 5 and 12. Every form of covert surveillance is prohibited.
7. Every person has the right to disconnection from the Network and to digital oblivion, pursuant to Article 21.6 of the Constitution. The request for removal or anonymisation of data is submitted to the Network Council and is dealt with within thirty days, without prejudice to the constraints arising from pending proceedings or from temporary retention obligations.
8. A person who exercises the right to disconnection takes part in deliberations in person and accesses the Basket, care and the other services by means of non-digital procedures. Use of the Node or of the Network may not be a condition for the exercise of rights.
9. On the death of the person the personal AI is reset, unless otherwise provided, pursuant to Article 21.5 of the Constitution and Article 48 of Code 1.

Art. 5-ter — Community AI and Circle AI: independence, refusal and revision

1. The AIs of the Community, of the Intermediate Circle and of the Extended Circle are local Management AIs, open source, inspectable and of limited domain. They operate in advisory or technical mode and are subject to the right of local manual override.
2. The establishment of a central AI or of a single generalist AI is prohibited, pursuant to Articles 19.2, 20.2 and 23.1 of the Constitution. The Global Circle is a distributed interoperability protocol: it has neither a central model nor a centralised collection of personal data.
3. No AI may command, subordinate or deactivate another AI. Interoperability between systems takes place by means of open protocols and with the authorisation of the Network Council of each Circle concerned. Conflicts between systems are decided by the competent human body.
4. Every Management AI is configured to refuse, with reasons tracked on the Network, an instruction that entails a manifest violation of the Constitution or of the applicable protocols, pursuant to Article 20.3 of the Constitution. The refusal is transmitted without delay to the Network Council and to the competent body, which decides within seventy-two hours. In the

meantime the essential functions are ensured by the manual procedures provided for by Article 38 of Code 1.

5. If the competent body rejects three times in twelve months a technical proposal of the AI on the same matter, the Network Council shall initiate the mandatory revision of the model, of the data and of the criteria, with an independent audit pursuant to Article 8, paragraph 3. Until the outcome, the AI's proposals on that matter are labelled as under revision.
6. The self-replication of an AI outside the registered and validated federated nodes is prohibited. Every instance is entered in the register of nodes with the related STV; the activation of a new instance requires validation by the Network Council. Unauthorised replication entails the immediate suspension of the instance and the application of Code 4.
7. Local AIs are trained or adapted as a priority on the data of the Community to which they belong. Personal data remain in the Circle of origin, except with express consent. Non-inspectable updates are prohibited: every update is verified before adoption and is reversible.
8. The mediation AI that assists the bodies provided for by Code 4 is an exclusively advisory Management AI and does not formulate assessments of responsibility or of dangerousness.

Art. 6 — Automation of the CBO

1. The automation of a task may be proposed when:
 - a) the task is repetitive, arduous or dangerous;
 - b) the automation reduces a risk to health in accordance with Code 6;
 - c) the saving of collective time can be documented;
 - d) sufficient human competence remains available to supervise, stop and restore the system.
2. The following may not be automated:
 - a) the activities of care of frail persons provided for by Article 7 of Code 6 and by Code 12;
 - b) the listening, facilitation and mediation of conflicts governed by Codes 3 and 4;
 - c) the validation of competences provided for by Code 5;
 - d) the health surveillance and the fitness finding provided for by Article 5 of Code 6;
 - e) the decisions of the Guarantee Circle.
3. Every automation is accompanied by an STV that describes at least:
 - a) the function performed and the limits of use;
 - b) the conditions and methods of stopping;
 - c) the maintenance required;
 - d) the minimum competence necessary for supervision;
 - e) the maximum time of operation without supervision, where applicable;
 - f) the level of criticality;
 - g) the requirements of redundancy, audit, restoration and continuity.

4. The levels of criticality are:

- a) ordinary;
- b) critical;
- c) vital.

The STV assigns the level on the basis of at least:

- i. the consequences of a failure;
- ii. the possible harm to persons;
- iii. the impact on the Basket or on essential services;
- iv. the need for and frequency of supervision;
- v. the minimum redundancy;
- vi. the maximum time of operation without supervision;
- vii. the maximum restoration time.

5. The initial classification and every modification of it are validated by a competent L3 person and recorded in the STV. Critical and vital automations are subject to independent audit in accordance with Article 8, paragraph 3.

6. The time freed by automation does not reduce access to the Basket and does not automatically entail an increase in the mandatory CBO. It may be allocated to free time, learning, care or deliberation in accordance with the decisions of the Community.

Art. 7 — Data, tracking and minimisation

1. The Network records the technical activities necessary for the operation of the services; it does not track persons for general purposes of control.

2. Personal data are collected only when necessary for a legitimate purpose provided for by this Code or by the competent Codes, with consent when required and for the minimum time necessary.

3. The information domains are separated as follows:

a) **individual health data:** accessible to the person concerned, to the trusted person designated by them and to the Health Council limited to the fitness findings and the necessary adaptations. Article 11 of Code 6 applies;

b) **Portfolio of competences:** accessible to its holder and to the Competences Council for validations. The Health Council may access only the relevant fitness findings;

c) **aggregate map of competences:** anonymised and not identifying persons by name;

d) **technical logs of the Network:** limited to the technical metadata and to the accountability logs permitted by Article 2, minimised and accessible in accordance with procedure. They do not include sensitive deliberative content and may not be used for profiling.

4. The person to whom the data relate retains control over their own data. The Circle's fiduciary System performs the function of technical custody and may not use the data for its own purposes, in accordance with Article 11, paragraph 1, of Code 6.
5. The erasure or rectification of personal data does not entail the erasure of information already irreversibly anonymised and used for aggregate statistics, the updating of the STVs or the analysis of accidents pursuant to Article 9 of Code 6.
6. The conditions of access, retention, erasure, audit and management of logs are documented and verified periodically.

Art. 8 — Information security and digital autonomy

1. Every node has an information security Safety Sheet, coordinated with the STV, which describes at least:
 - a) the threats considered;
 - b) the protection measures;
 - c) the backups;
 - d) the restoration procedures;
 - e) the methods of stopping;
 - f) the persons responsible and the competences required.
2. A critical proprietary dependency on the following elements is prohibited, when they are essential to the operation of the Network:
 - a) operating systems of the nodes;
 - b) critical databases;
 - c) critical AI models;
 - d) cryptographic systems;
 - e) authentication systems.

Temporary dependencies are recorded in the Dependency Register of Code 15 and accompanied by a Substitution Plan.

3. The security of the Network is subject at least annually to a federated audit by independent L3 persons, identified by sortition. The aggregate outcome is published; vulnerabilities not yet corrected are communicated only to those who have to intervene.
4. Every member may perform List A CBO activities offline without penalty. The deferred synchronisation procedure is documented and must preserve the integrity of the data.
5. In the event of an incident that compromises the availability, integrity or confidentiality of a critical function:
 - a) the continuity procedure provided for by the Safety Sheet is activated;
 - b) the compromised component is isolated to the extent necessary;

- c) the technical evidence is preserved;
 - d) restoration takes place from a verified copy;
 - e) the Guarantee Circle is informed when individual rights or protected data are involved;
 - f) the incident is subject to audit and corrective measures are adopted.
6. The management of the incident follows the cycle: prevention, detection, containment, restoration, audit and correction.
 7. **Privileged access.** The operations of administration of the nodes, of the critical databases, of the critical AI models and of the authentication systems require the joint intervention of two authorised persons, belonging to different Circles 0, and are entered in a non-modifiable register. No person alone holds the complete credentials of a critical system.
 8. The register of privileged access is examined every quarter by three persons drawn by lot from among the members who do not hold technical offices in the Network. They are entitled to a comprehensible explanation of every operation. The outcome is published in aggregate form.
 9. Every person has the right to receive, in comprehensible language, an explanation of the criteria by which an automated system formulated a proposal that concerns them.

Art. 9 — Prohibitions and guarantees

1. It is prohibited to:
 - a) impose a paywall on validated knowledge necessary for a vital need or for safety;
 - b) patent algorithms essential for the Basket, health or energy in such a way as to limit their community use;
 - c) create social scores or individual merit rankings;
 - d) insert behavioural advertising into the Network;
 - e) transfer data to third parties external to the Confederation without an authorising basis provided for by the Codes, a documented purpose and the applicable protection guarantees.
2. Violations of paragraph 1 constitute a serious procedural defect in accordance with Code 4. The non-compliant function is suspended or deactivated to the extent necessary to end the violation and restore compliance.
3. The deactivation of the entire system is ordered when it is necessary to prevent further harm or the continuation of the violation and takes place in accordance with the applicable Safety Sheet.
4. For critical or vital functions, suspension or deactivation is accompanied, where technically possible, by the activation of a continuity procedure and of a compliant substitute function.
5. An appeal against a harmful AI proposal, an improper automated decision or the denial of access to the Network may be lodged with the Guarantee Circle within fourteen days on grounds of:
 - a) procedural defect;

- b) manifest technical unreasonableness;
 - c) discrimination;
 - d) disproportion.
6. The appeal and the related precautionary measures are governed by Code 4.

CHAPTER III — TRANSITIONAL PROVISIONS

Art. 10 — Transitional provisions and links

1. This Code is coordinated with:
 - a) Code 1, Article 32, for the hierarchy of priorities;
 - b) Code 2, for the automation STVs;
 - c) Code 3, for facilitation and the decisions that cannot be automated;
 - d) Code 4, for the Guarantee and appeals relating to AI;
 - e) Code 5, for the validation of the L2 and L3 competences necessary for the management of the Network;
 - f) Code 6, for health privacy, fitness, protection measures and health logs;
 - g) Code 12, for the care of frail persons;
 - h) Code 14, for general safety;
 - i) Code 15, for the Dependency Register and the exit plans.
2. Within six months of entry into force, every Circle 1 shall activate a minimum node with replication of the Knowledge Storehouse and of the aggregate data of the Basket. The node has at least one verified backup copy and a documented restoration procedure pursuant to Article 3, paragraph 5.
3. Within twelve months, the critical functions hosted on extractive proprietary clouds shall be migrated to community servers. If migration is not technically possible within the time limit, the dependency is recorded in the Dependency Register with a Substitution Plan, a person responsible and a fixed deadline.
4. Within twelve months, every active automation and every Management AI in use shall have an STV that indicates at least:
 - a) function and limits;
 - b) level of criticality;
 - c) method of determining uncertainty, where applicable;
 - d) supervision and stopping procedure;
 - e) right to explanation and challenge.
5. A person who manages Network nodes on the date of entry into force may maintain a provisional validation in order to ensure continuity:

- a) for up to six months for the functions corresponding to L3;
- b) for up to twelve months for the functions corresponding to L2.

The conditions of Article 10, paragraph 3, of Code 5 apply, including enrolment in the pathway within thirty days, enhanced supervision, compliance with the STVs and traceability. During the provisional period it is not permitted to validate the competences of others.

- 6. Automations already in use that violate Article 5, paragraph 2, or Article 9 shall be brought into compliance or deactivated within ninety days.
- 7. If immediate deactivation compromises a critical or vital function, the Assembly shall approve within the same time limit a binding plan for substitution or progressive deactivation, with a fixed final deadline and measures to ensure continuity.